



A two-level coalition formation game for competitive swarms

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ABSTRACT

This paper investigates the coalition formation problem for police unmanned aerial vehicles (UAVs) to control malicious aerial targets within the context of public security issues. Research on coalition formation primarily focuses on intra-team collaboration. However, inter-team competition also exists, requiring each team to make decisions that account for mutual interactions. To address this problem, we propose a two-level coalition formation game model that integrates intra-team cooperation and inter-team competition. The Nash equilibrium of this game model represents the optimal coalition structure for both teams. However, solving the Nash equilibrium faces the challenge of dimensionality in large-scale scenarios with increasing UAV numbers. To address this issue, we prove that the coalition formation process within a team is a potential game and propose an inter-coalition and intra-coalition iterative (Double-IC) algorithm, embedding iterated best response within a fictitious play framework to effectively solve coalition structures in large-scale swarms. Finally, extensive numerical simulations were conducted to validate our proposed method in terms of equilibrium computation accuracy, computational time stability, decision-making robustness, and practical applicability in dynamic target assignment scenarios.

1. Introduction

1.1. Background and motivation

Unmanned aerial vehicles (UAVs) have been increasingly deployed for large-scale and complex tasks due to their cost efficiency, intrinsic maneuverability, and flexible deployment capabilities (Li et al., 2024; Liu et al., 2019; Qi et al., 2023; Ruan, Li, Cheng et al., 2022; Ruan, Li, Dai et al., 2022).

UAV swarms can accomplish more complex tasks through collaboration among individual agents, while the efficiency of task execution depends on the task planning of the UAVs. In coverage scenarios, UAV coverage and data transmission mechanism are analyzed and a multi-UAV cooperative deployment model is proposed to achieve better coverage performance (Li & Duan, 2017; Nemer et al., 2020). In dynamic planning scenarios, networked evolutionary game-theoretic approaches (Zhang et al., 2024) and multi-agent reinforcement learning-based approaches (Liu, Dou, Zhang, Zhang, & Zong, 2023) are used to study the dynamic task assignment problem of UAV swarms. Additionally, path planning (flight control) (He et al., 2021; Zhang et al., 2023) is also an important part of task planning. In this paper, we mainly

focus on the task assignment problem of UAV swarms aiming to achieve larger situation advantage and coverage rate.

However, misused and malicious UAVs have caused various problems such as disturbing airports and threatening public safety (Li, Ning, He, Lee, & Zhao, 2022). Motivated by these problems, we study the coalition formation (CF) of two competitive UAV swarms, where the police UAVs team and malicious aerial targets team compete with each other to achieve dominance through target assignment. Research on this task could be potentially applied in the terrorist attacks and social security issues in the future.

In competitive swarm scenarios, effective task assignment is crucial for the successful execution of objectives, which has garnered significant attention in recent years (Liles et al., 2023; Liu, Wang, Fu, & Wang, 2023; Zhang et al., 2021). Target assignment is a kind of task assignment where UAVs select their targets to maximize team benefits. In this process, UAVs selecting the same target autonomously form coalitions, collaborating to attack the target in the subsequent engagement (Wu et al., 2022), resulting in the partitioning of the swarm into multiple coalitions.

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The CF problem for UAVs has attracted a lot of attention, primarily formulated as multi-objective optimization models and game-theoretic models.

In actual swarm scenarios, there are typically multiple conflicting objectives to achieve, such as completing as many missions as possible with minimal resources. The multi-objective optimization framework provides a feasible technique to consider these conflicting objectives simultaneously. The cooperative mission assignment for heterogeneous UAVs is discussed in [Gao, Wang et al. \(2023\)](#), where a multi-objective optimization model is developed to balance mission gains and UAV losses. A multi-objective planning model is constructed to study dynamic mission planning for UAV formations ([Zhang et al., 2023](#)). Additionally, robot coalition formation is treated as a multi-objective optimization problem ([Agarwal et al., 2015](#)). Given the strategic interaction among UAVs, a game-theoretic approach provides an effective analytical tool ([Jang et al., 2018](#); [Pilloni et al., 2022](#)). Coalition formation is typically regarded as a classic cooperative scenario. In a cooperative scenario, a coalition formation game model is constructed to jointly optimize task selection and bandwidth allocation ([Chen et al., 2021](#)). Additionally, a Bayesian cooperative game is proposed to examine the coalition formation of the UAV wireless network, considering a distributed control setting under uncertainty ([Mittal et al., 2023](#)).

In real-world scenarios, the coalition structures of different teams mutually influence each other's intra-team coalition structures. Therefore, it is necessary to study coalition formation among different teams. In the field of physical layer security (PLS) in wireless communications, there has been extensive research on the problem of coalition formation among different groups. Specially, a novel three-party hierarchical game framework involving legitimate users (LUs), eavesdroppers (EVs), and jammers (JAs) is developed to study the dynamic trilateral coalitions of PLS ([Chen, Yi et al., 2024](#)). To address the unpredictability of wireless systems, a hierarchical game combining a matching subgame and a coalition formation subgame is formulated to model interactions among LUs, EVs, and third-party collaborative users (TCUs) ([Liu et al., 2025](#)). In the field of UAV swarms, a game-theoretical-based coalition formation algorithm ([Chen, Wei, & Duan, 2024](#)) is proposed to drive defenders to form different-sized defending coalitions. Focus on the battlefield environment, a multi-UAV dynamic mission planning algorithm based on the time window mechanism is proposed ([Zhang et al., 2023](#)) under a distributed formation structure.

These studies are conducted in the context of adversarial teams, where coalition formation considers the individual attributes of opposing teams and the constraints of their own resources. However, they do not fully account for the coalition structures of the opposing teams. Compared to research in wireless communications, studies on coalition formation in the context of UAV swarm cooperative confrontation, which simultaneously address intra-team and inter-team coalitions, are relatively scarce. One reason for this is the high dynamicity of individual agents, leading to real-time changes in the network topology, which further complicates the coupling relationship between inter-group and intra-group coalition structures. Therefore, research on this topic is both challenging and valuable.

The coalition formation problem is essentially a combinatorial optimization problem, and it has been proven to be an NP-hard problem ([Agarwal et al., 2015](#); [Li, Li, Vo, & Kowalczyk, 2022](#); [Sandholm et al., 1999](#); [Shehory & Kraus, 1995](#)). Currently, there are several methods available to solve coalition formation problems. Swarm intelligence methods are widely used in coalition formation, leading to robust and efficient solutions. Recent methods include particle swarm optimization-based algorithms ([Gao, Ma, Li, & Shen, 2023](#)), ant colony-based algorithms ([Wu et al., 2018](#)), wolf pack-based algorithms ([Lu et al., 2020](#)) and genetic-based algorithm ([Yan et al., 2024](#)). These methods are low cost and efficient, but suffer from a lack of guaranteed convergence, limited ability to adapt to dynamic environments, and a lack of effective coordination mechanisms, making them less reliable

and efficient for real-time, cooperative decision-making in complex scenarios.

Facing the complex and rapidly changing environment, reinforcement learning (RL) is also used to solve the decision-making problems for UAVs. RL offers numerous advantages, including handling uncertainties in the environment, real-time implementation, and dealing with different tasks ([Zhao et al., 2019](#)). Specifically, multi-agent reinforcement learning (MARL) has shown exceptional suitability and outstanding performance in UAV swarm decision-making. However, MARL faces significant limitations in multi-UAV task assignment due to the curse of dimensionality, challenges in generalizing learned policies to dynamic environments, and issues with stability and convergence, making it difficult to scale and apply to real-world scenarios. It can be observed that the coalition formation problem of multiple UAVs has not been adequately solved ([Chen, Wei, & Duan, 2024](#)). Therefore, it is essential to develop a novel algorithm that can address the limitations of traditional MARL and swarm intelligence algorithms to improve performance and efficiency in multi-UAV target assignment.

Motivated by the above observations, this paper studies the coalition formation problem for police UAVs to control malicious aerial targets within the context of public security issues. It constructs a novel coalition formation game model and proposes an algorithm to solve the equilibrium, representing the optimal coalition structures of both teams. Specifically, the main contributions of this paper are as follows:

- We construct a two-level coalition formation game model with a utility function that considers situation advantage and potential threats from unselected opponents. This model is composed of the inter-team zero-sum game model and the intra-team coordination game model.
- The stable coalition structure of the team is characterized as the equilibrium of the game model. We prove that the intra-team coordination game is a potential game with at least one pure Nash Equilibrium (NE).
- We propose the Double-IC algorithm, which is a novel distributed decision-making framework to solve the Nash equilibrium of game models. It embeds iterated best response within a fictitious play to effectively solve the coalition structure in large-scale swarm scenarios, and the convergence of the Double-IC algorithm is guaranteed.
- Extensive numerical experiments validate the algorithm's computational efficiency and stability through comparisons with the enumeration linear programming (ELP) algorithm. Additionally, the robustness of our method is confirmed via comprehensive sensitivity analyses and benchmarking against state-of-the-art reinforcement learning algorithms. Finally, the practical applicability of the Double-IC algorithm is demonstrated through its deployment in dynamic target assignment scenarios within UAV swarm operations.

The remainder of this paper is organized as follows. In Section 2, we provide the problem description. Section 3 details the evaluation of the situation and motion control for UAV swarms. Section 4 introduces the team utility and individual utility. In Section 5, we present the potential game approach to intra team coalition formation and prove that it is a potential game. Section 6 proposes the Double-IC algorithm. Numerical results and performance analyses are conducted in Section 7. Finally, Section 8 concludes the paper.

2. Problem formulation

We consider a competitive UAV swarm scenario with N police UAVs forming the red team, denoted by $R = \{1, 2, \dots, N\}$, and M malicious aerial targets forming the blue team, denoted by $B = \{1, 2, \dots, M\}$. It is assumed that each UAV can target one opponent at a time, operates independently within the swarm, and maintains full communication with other UAVs in the team.

Definition 1 (Team Strategy). The strategies of the red and blue teams are binary matrices denoted by X and Y , respectively, as shown in Eqs. (1) and (2).

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,M} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,M} \end{bmatrix}, \quad (1)$$

$$Y = \begin{bmatrix} y_{1,1} & y_{1,2} & \cdots & y_{1,N} \\ y_{2,1} & y_{2,2} & \cdots & y_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ y_{M,1} & y_{M,2} & \cdots & y_{M,N} \end{bmatrix}. \quad (2)$$

$x_{i,j}$ is the decision variable for UAV i from the red team. If UAV j from the blue team is assigned to i , then $x_{i,j} = 1$, otherwise, $x_{i,j} = 0$. We can define $y_{j,i}$ in the same way.

Based on the definition of team strategy, we can give the definition of individual strategy and coalition as follows.

Definition 2 (Individual Strategy). The strategy of UAV i of red team is a binary vector $s_i^R = [x_{i,1}, x_{i,2}, \dots, x_{i,M}]$, $i \in R$, such that $\sum_{j \in B} x_{i,j} \leq 1$, $x_{i,j} \in \{0, 1\}$, and the strategy of UAV j of blue team is a binary vector $s_j^B = [y_{j,1}, y_{j,2}, \dots, y_{j,N}]$, $j \in B$, such that $\sum_{i \in R} y_{j,i} \leq 1$, $y_{j,i} \in \{0, 1\}$.

Definition 3 (Coalition). A coalition of the red team is a UAV set such that $C_j = \{i \mid x_{i,j} = 1, i \in R\}$, and a coalition of the blue team is a UAV set such that $C_i = \{j \mid y_{j,i} = 1, j \in B\}$.

The key to coalition formation problems is determining the quality of given coalition structures. This paper primarily quantifies coalition structures based on two factors: situation advantage and potential threat.

The situation advantage reflects the offensive and defensive positioning of the UAV relative to its selected target. The situation advantage matrices of the red team and the blue team are denoted by F_R and F_B , respectively, as presented in Eqs. (3) and (4).

$$F_R = \begin{bmatrix} F_{1,1}^r & F_{1,2}^r & \cdots & F_{1,M}^r \\ F_{2,1}^r & F_{2,2}^r & \cdots & F_{2,M}^r \\ \vdots & \vdots & \ddots & \vdots \\ F_{N,1}^r & F_{N,2}^r & \cdots & F_{N,M}^r \end{bmatrix}, \quad (3)$$

$$F_B = \begin{bmatrix} F_{1,1}^b & F_{1,2}^b & \cdots & F_{1,N}^b \\ F_{2,1}^b & F_{2,2}^b & \cdots & F_{2,N}^b \\ \vdots & \vdots & \ddots & \vdots \\ F_{M,1}^b & F_{M,2}^b & \cdots & F_{M,N}^b \end{bmatrix}. \quad (4)$$

The element $F_{i,j}^r$ in F_R is the situation advantage of UAV i of the red team relative to its target j . The definitions of $F_{i,j}^r$ and $F_{j,i}^b$ are provided in Sections 3 and 4, respectively.

For the situation advantage of a coalition, we consider not only the situation advantage relative to the target but also the weakening effect of the opponent's situation advantage against the team members. Potential threat is caused by the opponent's UAVs that remain uncovered. It is noted that the potential threat defined in this paper considers not only the rate of uncovered UAVs but also the interactive effects of potential threats from both teams. The potential threat to the red team and blue team are denoted by T_R and T_B , respectively. Detailed definitions are provided in Section 4.

In this paper, we develop a two-level coalition formation model comprising an inter-team zero-sum game and an intra-team coordination game.

Firstly, we give the Inter-team zero-sum game model, treating the red and blue teams each as a single agent to characterize their adversarial relationship.

Definition 4 (Inter-Team Zero-Sum Game Model). An inter-team zero-sum game model Γ is a tuple $\langle \{R, B\}, S, \{U_R, U_B\} \rangle$, where

- $\{R, B\}$ is a set of UAVs for the red team and blue team;
- $S = \{X, Y\}$ is the strategy space of the UAV swarms, where X and Y are the feasible sets of decision matrices for the red team and blue team, respectively, and we have $X \in X$ and $Y \in Y$;
- U_R and U_B refer to the team utility functions for the red team and blue team, respectively, and $U_R(X, Y) = -U_B(X, Y)$. The team utility for red team is defined as follows.

$$U_R = [S_R(F_R, F_B, X, Y) - T_R(X, Y) - [S_B(F_R, F_B, X, Y) - T_B(X, Y)], \quad (5)$$

where S_R and S_B are functions correlated with F_R and F_B , representing the situation advantage of the red and blue teams, respectively. The detailed definitions of S_R and S_B are shown in Eqs. (22) and (23), respectively.

Secondly, we give the intra-team coordination game model to analyze the cooperative relationship among the UAVs within a team.

Definition 5 (Intra-Team Coordination Game Model). An intra-team coordination game model Γ is a tuple $\langle \alpha, P^\alpha, S^\alpha, U_j^\alpha, j \in P^\alpha \rangle$, where

- α is the role variable, $\alpha = \text{red}$ and $\alpha = \text{blue}$ indicate that Γ is the intra-team coordination game model for the red team and blue team, respectively.
- P^α is the set of UAVs for the corresponding team. Specifically, $P^\alpha = \{1, 2, \dots, N\}$ when $\alpha = \text{red}$, and $P^\alpha = \{1, 2, \dots, M\}$ when $\alpha = \text{blue}$.
- $S_j^\alpha, j \in P^\alpha$ is the strategy space of the UAV j from the α team, denoted as $S_j^\alpha = \{s_j^\alpha \mid \sum_{k \in P^\alpha} s_{j,k}^\alpha \leq 1, s_{j,k}^\alpha \in \{0, 1\}, \forall j \in P^\alpha\}$, where $\bar{P}^\alpha$ denotes the set of UAVs belonging to the opposing team. s_j^α refers to the individual strategy in Definition 2. $s_{j,k}^{\text{red}}$ refers to $x_{j,k}$ for $j \in R$ and $k \in B$, while $s_{j,k}^{\text{blue}}$ refers to $y_{j,k}$ for $j \in B$ and $k \in R$.
- $U_j^\alpha(s_j^\alpha, s_{-j}^\alpha), j \in P^\alpha$ is the individual utility function of α team, where s_{-j}^α denotes the strategy set of all UAVs in team α except for UAV j . The individual utility function in Definition 5 is defined in Section 4.2.

Finally, the optimal coalition structures of UAV teams can be characterized as the Nash equilibrium obtained by solving the aforementioned game models. The equilibrium in coalition structures of competitive UAV swarms has significant practical implications. First, it ensures a stable and predictable operational state where no coalition has the incentive to unilaterally change its strategy, which is crucial for mission planning and resource allocation. Furthermore, in scenarios where the opposing swarm's strategies are uncertain, achieving equilibrium enables each coalition to secure the maximum possible utility regardless of the opponent's actions, thus providing a robust decision-making framework for real-world adversarial environments. The basic definitions of coalition structure and Nash equilibrium are given as follows.

Definition 6 (Coalition Structure). A coalition structure $\pi = \{C_1, C_2, \dots, C_n\}$ is a disjoint partition of $\mathcal{N}$ such that $\cup_{i=1}^n C_i = \mathcal{N}$ and $C_i \cap C_j = \emptyset$ for $\forall i, j \in \mathcal{N}$ and $i \neq j$.

Definition 7 (Nash Equilibrium). A coalition structure $\pi^* = \{C_1^*, C_2^*, \dots, C_n^*\}$ is a pure-strategy Nash equilibrium if and only if no player can improve its utility by changing its choice while others remain their selection, that is,

$$U_i(c_i^*, c_{-i}^*) \geq U_i(c_i, c_{-i}^*), \forall i \in S, \forall c_i \in A_i, c_i \neq c_i^*. \quad (6)$$

In this section, we characterize the coalition formation problem of UAV swarms using mathematical models, outlining the basic framework and relevant definitions for studying this issue. In the next section, we will define the situation advantage function for one-on-one scenarios.

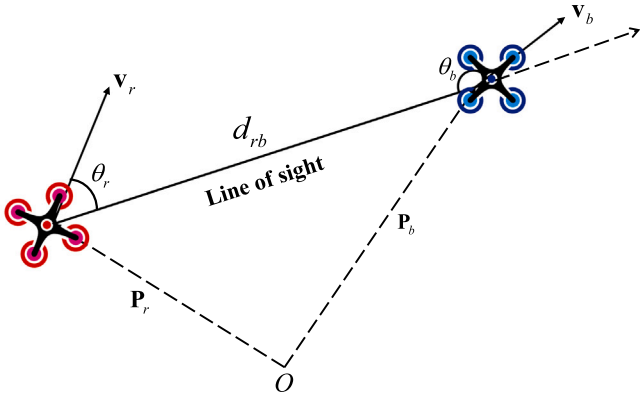


Fig. 1. Schematic of relative angle and distance.

3. The evaluation of situation and motion control for UAV swarms

3.1. Relative situation of UAVs in one-on-one scenario

The position relationship of the one-on-one scenario is shown in Fig. 1.

In this figure, $d_{i,j}$ is the relative distance between i and j .

$$d_{i,j} = \|\mathbf{P}_i - \mathbf{P}_j\| = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (7)$$

The line of sight refers to the direction pointing from itself to the opponent. θ_{ij} is the included angle between the direction of the red's velocity and line of sight, named line-of-sight angle of i .

$$\theta_{ij} = \arccos \frac{\mathbf{v}_i \cdot (\mathbf{P}_j - \mathbf{P}_i)}{\|\mathbf{v}_i\| \|\mathbf{P}_j - \mathbf{P}_i\|}, \quad (8)$$

$$\theta_{ji} = \arccos \frac{\mathbf{v}_j \cdot (\mathbf{P}_i - \mathbf{P}_j)}{\|\mathbf{v}_j\| \|\mathbf{P}_i - \mathbf{P}_j\|},$$

where the symbol $\cdot$ represents the inner product of two vectors, $\|\cdot\|$ is the modulus of a vector. v_i and v_j are the velocity vectors of i and j , respectively.

The capture zone C_i of UAV i can be defined based on the relative distance and the line-of-sight angle. A target UAV will be captured when it enters this capture zone.

$$C_i = \{(d_{ij}, \theta_{ij}) \in \mathbb{R}^+ \times [0, \pi] \mid d_{ij} \leq d_c, \theta_{ij} \leq \theta_c\}, \quad (9)$$

where d_c is the capture distance threshold, θ_c is the capture angle threshold.

3.2. Situation evaluation in one-on-one scenario

In this section, we will evaluate the situation advantage in one-on-one scenario based on the line-of-sight angle and distance between UAVs.

3.2.1. Angle advantage function

In 3-dimensional space, we mainly use the angle between the sight angle and the velocity to describe the angle advantage of the UAV. In this paper, the angle advantage function (Lyu et al., 2022) of i is denoted by

$$F_{i,j}^a = 1 - \frac{\theta_i + (\pi - \theta_j)}{2\pi}. \quad (10)$$

Similarly, we can get the angle advantage of j as follows:

$$F_{j,i}^a = 1 - \frac{(\pi - \theta_i) + \theta_j}{2\pi}. \quad (11)$$

Obviously, we have $F_i^a + F_j^a = 1$.

3.2.2. Distance advantage function

In real scenarios, the distance advantage is related to the angle advantage. We give the definition of distance advantage of i as follows.

$$F_{i,j}^d = \begin{cases} e^{-\frac{|d_{i,j} - D_{\min}|}{\max\{d_{i,j} - D_{\min}, \delta\}}}, & F_{i,j}^a \geq F_{j,i}^a \\ e^{-\frac{|D_{\max} - D_{\min}|}{\max\{d_{i,j} - D_{\min}, \delta\}}}, & F_{i,j}^a < F_{j,i}^a \end{cases} \quad (12)$$

where $D_{\min}$ is the minimum relative distance with the opponent to ensure the security. δ is a very small positive number, ensuring the denominator does not equal 0. The distance advantage function of j , $F_{j,i}^d$, can be defined in the same way.

If i has the superior angle, the distance advantage increases as $d_{i,j}$ approaches $D_{\min}$, indicating an offensive situation. The denominator $\max\{d_{i,j} - D_{\min}, \delta\}$ ensures $d_{i,j}$ remains above the security distance $D_{\min}$, as the advantage decreases if $d_{i,j} \leq D_{\min}$. Conversely, if i is in a defensive situation with a smaller angle advantage than j , the distance advantage increases with the difference between $d_{i,j}$ and $D_{\min}$.

3.2.3. Situation advantage function

In this paper, the situation advantage is evaluated by the weighted sum of different advantage factors, with the weights reflecting their importance. The situation advantage function of i is given as follows.

$$F_{i,j} = \omega_1 F_{i,j}^d + \omega_2 F_{i,j}^a$$

$$\begin{cases} \omega_1 + \omega_2 = 1, \\ \omega_1 < \omega_2, & \text{if } d_{i,j} \leq D_{\min} \\ \omega_1 > \omega_2, & \text{if } d_{i,j} \geq D_{\max} \\ \omega_1 = \omega_2 = 0.5, & \text{else} \end{cases} \quad (13)$$

where ω_1 and ω_2 represent weights assigned to distance advantage and angle advantage, respectively, which can be adapted to situations.

3.2.4. Motion control for UAV swarms

The motion control strategy for UAV swarms combines target pursuit with threat avoidance to achieve effective combat maneuvers. At time t , the control system calculates desired movement vector $\mathbf{v}(t)$ for each UAV by considering three velocity vectors: the target pursuit vector $\mathbf{v}_{tp}(t)$, the target threat avoidance vector $\mathbf{v}_{tt}(t)$ and the other avoidance vector $\mathbf{v}_{ot}(t)$.

$$\mathbf{v}(t) = w_{tp}(t)\mathbf{v}_{tp}(t) + w_{tt}(t)\mathbf{v}_{tt}(t) + w_{ot}(t)\mathbf{v}_{ot}(t), \quad (14)$$

where $w_{tp}(t)$, $w_{tt}(t)$, $w_{ot}(t)$ are the corresponding weights.

The target pursuit vector $\mathbf{v}_{tp}(t)$ is designed as the normalized direction toward the target, enabling efficient interception:

$$\mathbf{v}_{target}(t) = \frac{\mathbf{p}_{target}(t) - \mathbf{p}(t)}{\|\mathbf{p}_{target}(t) - \mathbf{p}(t)\|}, \quad (15)$$

where $\mathbf{p}_{target}(t)$ and $\mathbf{p}(t)$ denote the positions of the target UAV and the current UAV respectively.

For target threat avoidance, we propose a perpendicular evasion strategy to maintain pursuit while avoiding direct confrontation:

$$\mathbf{v}_{tt}(t) = \text{sign}(\mathbf{v}_{rel}(t) \cdot \mathbf{n}(t))\mathbf{n}(t), \quad (16)$$

where $\mathbf{v}_{rel}(t)$ is the relative velocity to the target, $\mathbf{n}(t)$ is the unit normal vector to the target direction. This design enables the UAV to maintain pursuit while staying outside the target's capture zone.

The other-threat avoidance vector incorporates both distance and angular relationships to generate effective evasive maneuvers:

$$\mathbf{v}_{ot}(t) = - \sum_{j \in \mathcal{T}} \eta_j(t) \frac{\mathbf{p}_j(t) - \mathbf{p}(t)}{\|\mathbf{p}_j(t) - \mathbf{p}(t)\|}, \quad (17)$$

where $\mathcal{T}$ represents the set of threatening UAVs.

The threat level $\eta_j(t)$ is evaluated using a composite function that considers both proximity and angular aspects:

$$\eta_j(t) = k_1 e^{-\|p_j(t)-p(t)\|/d_0} + k_2 e^{-\theta_j(t)/\theta_0}, \quad (18)$$

where k_1 and k_2 are weighting coefficients, d_0 and θ_0 are characteristic distance and angle parameters respectively.

4. Team utility and individual utility

In the following contents, the red team and blue team will be partitioned into a set of disjoint coalitions as $\pi = \{C_1, C_2, \dots, C_n\}$ based on Eq. (5).

4.1. Team utility

For the convenience of following analysis, the team utility of red team U_R in Eq. (5) is given by

$$\begin{aligned} U_R(X, Y) &= [S_R(F_R, F_B, X, Y) - S_B(F_R, F_B, X, Y)] - \\ &\quad [T_R(X, Y) - T_B(X, Y)] \\ &= U_{SR} - U_{TR}. \end{aligned} \quad (19)$$

Similarly, we can transform the team utility of blue team U_B . Next, we will introduce the comprehensive situation advantage (S_R and S_B) and the potential threat (T_R and T_B).

4.1.1. Team advantage

Before introducing the team advantage, we define the coalition advantage. Specifically, given a coalition $C_j, j \in B$ of the red team, the coalition advantage of which is given by

$$\begin{aligned} S_R^{C_j} &= \sum_{i \in R} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b) \\ &= \sum_{i \in C_j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b). \end{aligned} \quad (20)$$

For each UAV i in coalition C_j , the target j is selected, resulting in $x_{i,j} = 1$ for $i \in C_j$ in the second equation. Then the coalition advantage without reduction is $\sum_{i \in R} x_{i,j} F_{i,j}^r$. So the team advantage without reduction of the red side is given as follows:

$$\hat{S}_R(F_R, X) = \sum_{j \in B} \sum_{i \in R} x_{i,j} F_{i,j}^r. \quad (21)$$

However, this advantage will be reduced if the members of team are targeted by the opponent. While a higher $F_{k,i}^b$ indicates a greater reduction in the coalition advantage, thereby decreasing the remaining advantage of coalition C_j . This reduction factor is denoted by $1 - y_{k,i} F_{k,i}^b$. Given that these effects are interdependent, we quantify the cumulative reduction of the coalition advantage using a multiplicative approach, represented by $\prod_{k \in B} (1 - y_{k,i} F_{k,i}^b)$.

In conclusion, the team advantage $S_R(F_R, F_B, X, Y)$ of the red side is given by:

$$S_R(F_R, F_B, X, Y) = \sum_{j \in B} S_R^{C_j} = \sum_{j \in B} \sum_{i \in R} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b). \quad (22)$$

Similarly, the team advantage $S_B(F_R, F_B, X, Y)$ of the blue side is given as follows:

$$S_B(F_R, F_B, X, Y) = \sum_{j \in B} \sum_{i \in R} y_{j,i} F_{j,i}^b \prod_{k \in R} (1 - x_{i,k} F_{i,k}^r). \quad (23)$$

4.1.2. The potential threat to one team

The potential threat to the team mainly depends on the number of selected target of each team. Considering the interactive influence between two teams, the potential threat to the red team is given by

$$T_R(X, Y) = (1 - \frac{m}{M}) \frac{n}{N}, \quad (24)$$

Table 1

Special cases of potential threat to two teams.

	m	n	T_R	T_B
case1	M	N	0	0
case2	0	N	1	0
case3	M	0	0	1

where $m = \sum_{j \in B} \rho_j$ is the number of coalitions of the red team, and $n = \sum_{i \in R} \rho_i$ is the number of coalitions of the blue team. ρ_j and ρ_i are defined as follows:

$$\rho_j = \begin{cases} 1, & \text{if } \sum_{i \in R} x_{i,j} \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

$$\rho_i = \begin{cases} 1, & \text{if } \sum_{j \in B} y_{j,i} \neq 0 \\ 0, & \text{otherwise} \end{cases} \quad (26)$$

In Eq. (24), $\frac{m}{M}$ is the proportion of blue UAVs assigned to the red team. $1 - \frac{m}{M}$ is the proportion of blue UAVs that not assigned to the red team, with these unassigned UAVs posing a potential threat to the red team.

However, this potential threat will be reduced because of the unassigned red UAVs to the blue team. The higher the proportion of red UAVs that are not selected as targets (i.e. $1 - \frac{n}{N}$), the less potential threat the blue team poses to the red team (i.e. $1 - \frac{m}{M}$). In other words, $1 - \frac{m}{M}$ is directly proportional to $\frac{n}{N}$. Accordingly, T_R is defined as Eq. (24). Similarly, the potential threat to the blue team is given by

$$T_B(X, Y) = (1 - \frac{n}{N}) \frac{m}{M}. \quad (27)$$

Table 1 displays three special cases of potential threat to both teams. Case 1 represents the fully covering situation for both teams, resulting in zero potential threat to either team. Note that this case holds only when $M = N$. In Case 2, the blue team fully covers the red team without the red team forming any coalitions. Consequently, the blue team does not face any potential threat from the red team, while the red team is under maximum potential threat. Similarly, in Case 3, the red team fully covers the blue team without the blue team forming any coalitions. Hence, the red team does not face any potential threat from the blue team, while the blue team is under maximum potential threat.

4.2. Individual utility

Similar to Eq. (19), the individual utility of the red team is given by

$$U_R^i = U_{SR}^i - U_{TR}^i. \quad (28)$$

4.2.1. Individual advantage

The individual advantage of UAV i from the red team denoted by U_{SR}^i is defined as follows:

$$\begin{aligned} U_{SR}^i &= \sum_{b \in B} \sum_{r \in R} y_{b,r} F_{b,r}^b x_{i,b} F_{i,b}^r \prod_{r \in R, r \neq i} (1 - x_{r,b} F_{r,b}^r) \\ &\quad + \sum_{j \in B} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b) \\ &\quad - \sum_{j \in B} y_{j,i} F_{j,i}^b \prod_{k \in R} (1 - x_{i,k} F_{i,k}^r). \end{aligned} \quad (29)$$

The first part refers to the team advantage of the blue side reduced by UAV i . The second part represents the increase in the team advantage of the red side attributable to UAV i . The third part represents the increase in the team advantage of the blue side attributable to the presence of UAV i , which can also be understood as the potential risk posed to the red team when UAV i is targeted.

In the first part, $\sum_{b \in B} \sum_{r \in R} y_{b,r} F_{b,r}^b$ is the total advantage of the blue team, and $\sum_{b \in B} \sum_{r \in R} y_{b,r} F_{b,r}^b \prod_{r \in R, r \neq i} (1 - x_{r,b} F_{r,b}^r)$ reflects the team advantage of the blue side after cumulative reducing by UAVs from

the red team excluding i . For the purpose of clarity, it can be denoted by Q . Obviously, $Q(1 - \sum_{b \in B} x_{i,b} F_{i,b}^r)$ equals the advantage of the blue team in Eq. (23). Hence, $Q - Q(1 - \sum_{b \in B} x_{i,b} F_{i,b}^r) = Q \sum_{b \in B} x_{i,b} F_{i,b}^r$ is the team advantage of the blue side reduced by i , i.e., the first part of Eq. (29).

In the second part, $\sum_{j \in B} x_{i,j} F_{i,j}^r$ is the situation advantage of i without considering the hurt from the blue team. $\prod_{k \in B} (1 - y_{k,i} F_{k,i}^b)$ is the cumulative weakening from the red team. Accordingly, $\sum_{j \in B} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b)$ is the actual situation advantage of i .

In the third part, if i is not targeted, we can get $y_{j,i}^b = 0, j \in B$, that is, the third part equals 0. Otherwise, the third part does not equal 0.

4.2.2. The potential threat to individual

The potential threat to the individual i of the red team is denoted by U_{TR}^i . In this paper, U_{TR}^i is designed as its marginal contribution to U_{TR} , which takes the form

$$\begin{aligned} U_{TR}^i &= [T_R(X, Y) - T_B(X, Y)] - [T_R(X_{-i}, Y) - T_B(X_{-i}, Y)] \\ &= \left[\left(1 - \frac{m}{M}\right) \frac{n}{N} - \left(1 - \frac{n}{N}\right) \frac{m}{M} \right] \\ &\quad - \left[\left(1 - \frac{m'}{M}\right) \frac{n}{N} - \left(1 - \frac{n}{N}\right) \frac{m'}{M} \right], \end{aligned} \quad (30)$$

where X_{-i} represents the target selections of other UAVs of the red team except UAV i , and m' represents the number of coalitions without i . $T_R(X_{-i})$ refers to the threat to the red team without UAV i .

It is worth noting that there are two situations for the value of m' . If the coalition of i has more than one member, i.e. $|C_j| > 1$, then we have $m' = m$. If the coalition of i has only one member, i.e. $C_j = \{i\}$, then we have $m' = m - 1$.

4.3. Enumeration linear programming algorithm

Combining Eq. (5), Eq. (22), Eq. (23), Eq. (24), and Eq. (27), we have the bilateral optimization problem of coalition formation between two teams.

$$\begin{aligned} \max_{X \in \mathcal{X}} \min_{Y \in \mathcal{Y}} & \left[\sum_{j \in B} \sum_{i \in R} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b) - \left(1 - \frac{m}{M}\right) \frac{n}{N} \right] \\ & - \left[\sum_{j \in B} \sum_{i \in R} y_{j,i} F_{j,i}^b \prod_{k \in R} (1 - x_{i,k} F_{i,k}^r) - \left(1 - \frac{n}{N}\right) \frac{m}{M} \right] \\ \text{s.t.} & \begin{cases} x_{i,j}, y_{j,i} \in \{0, 1\}, \\ \sum_{j \in B} x_{i,j} \leq 1, \quad i = 1, 2, \dots, N, \\ \sum_{i \in R} y_{j,i} \leq 1, \quad j = 1, 2, \dots, M. \end{cases} \end{aligned} \quad (31)$$

The first constraint defines the permissible values for the decision variables of both teams. The second and third constraints ensure that each UAV selects at most one target.

To solve Eq. (31), a common approach is to first transform the game model into a matrix game by enumerating all strategy profiles of the UAVs from two teams. We call this approach enumeration linear programming (ELP) algorithm. Here, we present an example to show the computation details of ELP algorithm.

Example 1. In a public security scenario, a blue team of UAVs attempts to infiltrate an airport while being intercepted by a red defensive team, consisting of five UAVs ($N = 5$) and four UAVs ($M = 4$) respectively. Both teams must perform target assignment by considering their competitive advantages against opponents and coverage rates. The strategy space encompasses $4^5 = 1024$ possible assignments for the red team and $5^4 = 625$ assignments for the blue team, resulting in a total of $1024 \times 625 = 640000$ strategy profiles. Table 2 presents the situation advantage metrics, where each element ($F_{i,j}^r, F_{j,i}^b$) represents the respective situation advantages of red team UAV R_i and blue team UAV B_j during their engagement.

Table 2
Situation advantage of each UAV.

	B_1	B_2	B_3	B_4
R_1	(0.5241, 0.2273)	(0.4590, 0.3488)	(0.7167, 0.1393)	(0.1488, 0.5224)
R_2	(0.3636, 0.4489)	(0.3375, 0.4462)	(0.3612, 0.4856)	(0.2824, 0.5257)
R_3	(0.4917, 0.2491)	(0.4959, 0.2320)	(0.2717, 0.4638)	(0.2627, 0.4829)
R_4	(0.4775, 0.3557)	(0.4644, 0.3406)	(0.4556, 0.4126)	(0.3702, 0.4566)
R_5	(0.2733, 0.4755)	(0.2519, 0.4632)	(0.3371, 0.4859)	(0.2528, 0.5396)

Given the strategy profiles are extensive, we set an example to illustrate the calculation details of team utility for a given strategy profile, where the red team forms four coalitions, denoted by $\pi_R = \{C_1, C_2, C_3, C_4\} = \{\{R_1, R_2\}, \{R_3\}, \{R_4\}, \{R_5\}\}$ and the blue team form three coalitions, denoted by $\pi_B = \{C_1, C_2, C_3, C_4, C_5\} = \{\emptyset, \{B_2\}, \{B_3, B_4\}, \{B_1\}, \emptyset\}$. C_k denote the members in this coalition all select k from the opposing team as their target.

Fig. 2 displays the coalition structure of above example. Given the situation advantage of each UAV in Table 2, we can calculate the comprehensive situation advantage of C_1 and C_2 as follows:

$$S_R^{C_1} = F_{1,1}^r (1 - F_{3,1}^b) + F_{2,1}^r (1 - F_{2,2}^b) = 0.6525;$$

$$S_R^{C_2} = F_{3,2}^r (1 - F_{3,3}^b) (1 - F_{4,3}^b) = 0.1375.$$

Similarly, we have $S_R^{C_3} = 0.2936, S_R^{C_4} = 0.2528, S_B^{C_2} = 0.2249, S_B^{C_3} = 0.6133, S_B^{C_4} = 0.1077$. Then the comprehensive situation advantages and potential threats of each team are calculated as follows:

$$S_R = S_R^{C_1} + S_R^{C_2} + S_R^{C_3} + S_R^{C_4} = 1.3364, \quad T_R = (1 - \frac{4}{5}) \frac{3}{5} = 0;$$

$$S_B = S_B^{C_2} + S_B^{C_3} + S_B^{C_4} = 0.9459, \quad T_B = (1 - \frac{3}{5}) \frac{4}{5} = \frac{2}{5}.$$

Similarly, we can calculate the team utilities for both teams across other strategy profiles. We employ Gurobi, a high-performance mathematical programming solver, to solve the optimal coalition structures of both teams based on the team utility matrices. The results are given as follows:

$$\pi_R^* = \{C_1, C_2, C_3, C_4\} = \{\{R_2\}, \{R_3\}, \{R_1, R_4\}, \{R_5\}\};$$

$$\pi_B^* = \{C_2, C_3, C_4, C_5\} = \{\{B_2\}, \{B_3\}, \{B_1\}, \{B_4\}\}.$$

The ELP algorithm requires 500 s for computation to get the above optimal coalition structures. Under this example, the size of team utility matrix for the red team is $4^5 \times 5^4$ and $5^4 \times 4^5$ for the blue team. Obviously, this algorithm suffers an exponential increase in the number of strategy profiles with the increase in UAV numbers, which faces significant challenges for solution in large-scale scenarios with a time complexity of $O(M^N \cdot N^M)$. N and M are the number of UAVs for the red team and blue team, respectively. Hence, it is evident that using the ELP algorithm is impractical for the coalition formation problem in large-scale scenarios.

To avoid this exponential complexity, in Section 5, we demonstrate that the intra-team coalition formation game is a potential game, leading to the decomposition of the strategy optimization problem of one team into individual strategy optimization subproblems.

5. A potential game approach to intra-team coalition formation

Potential games are an important concept in game theory, it plays a prominent role in the cooperative control of distributed multi-agent systems, stemming from the fact that it guarantees each player's localized utility is aligned with the global objective (Li & Duan, 2017; Marden et al., 2009; Monderer & Shapley, 1996). Therefore, we only need to optimize each individual's strategy independently, without considering the team's entire strategy matrix. This approach effectively avoids the exponential growth problem of the strategy space and strategy

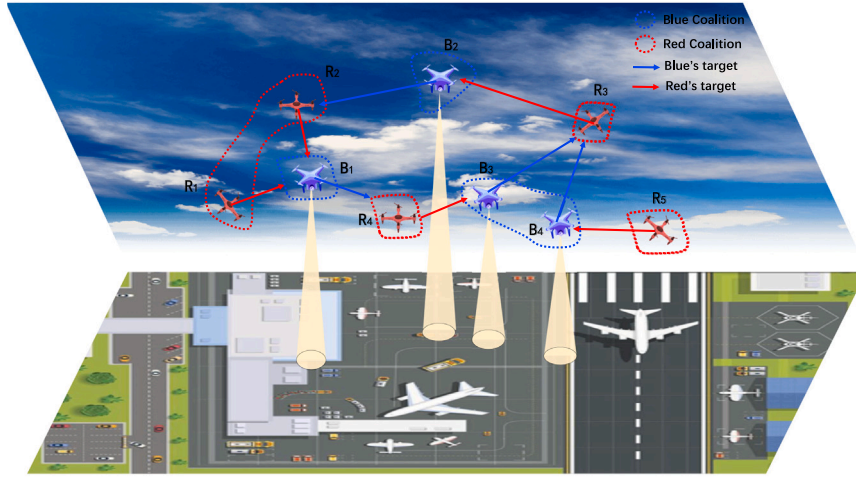


Fig. 2. Schematic diagram of target assignment example.

profiles in the ELP algorithm. In this section, we prove that the intra-team coordination game model defined in Definition 5 is a potential game, thereby achieving the decomposition of the coalition formation problem.

5.1. Basic definition of potential game

Firstly, we introduce some basic definitions about potential games and Nash equilibrium.

Definition 8 (Exact Potential Games). A game denoted by $G(S, \{A_i, i \in S\}, \{U_i, i \in S\})$ is called an exact potential game if there exists a global function $\Phi : A \rightarrow \mathbf{R}$ such that,

$$U_i(a_i, a_{-i}) - U_i(a'_i, a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a'_i, a_{-i}), \quad \forall i \in S, a_i, a'_i \in A_i, a_{-i} \in A_{-i}, \quad (32)$$

where the global function Φ is known as the potential function of game G .

In the following contents, we will refer to exact potential games simply as potential games. In potential games, the potential function captures the collective incentives of all players to adjust their strategies. When a UAV switches coalitions, its individual gain directly reflects the change in the potential function. For instance, if the potential function represents the total area surveilled, a UAV joining the coalition that maximizes its coverage area will simultaneously increase its own reward and the team's overall surveillance capacity. This ensures that decentralized UAVs, acting without explicit coordination, naturally optimize global efficiency while minimizing communication overhead. This property makes potential games particularly suitable for modeling large-scale UAV swarm coordination, as they balance individual autonomy with global efficiency. Additionally, (Monderer & Shapley, 1996) proves that a potential game possesses at least one pure Nash equilibrium. Thus, the existence of a Nash equilibrium is theoretically guaranteed, ensuring stable and predictable coalition structures.

5.2. Coalition formation of intra-team formulated as a potential game

In essence, coalitional games involve a set of players, denoted by $\mathcal{N} = \{1, 2, \dots, N\}$ who seek to form cooperative groups, i.e., coalitions, to strengthen their positions in the game (Saad et al., 2009). In this paper, we employ potential game theory to study the coalition partitioning problem of UAV swarm.

Lemma 1. Given the coalition structure of the blue team, the target selection problem of the red team can be formulated as a multi-player game, where $R = \{1, 2, \dots, N\}$ is the set of players, and $X = x_1 \times x_2 \times \dots \times x_n$ is the set of joint actions. $x_i = j$ refers to UAV i selecting UAV j of the blue team as its target, i.e., UAV i belongs to coalition c_j . If UAV i takes U_{TR}^i defined by Eq. (30) as the individual threat, then it constitutes a potential game with the potential function U_{TR} defined in Eq. (19).

Proof. Let x_i and x'_i be two possible target selection actions for UAV i , and let x_{-i} denote joint target selection actions of the remaining UAVs. According to the definition of individual threat in Eq. (30), we get

$$\begin{aligned} & U_{TR}^i(x_i, x_{-i}) - U_{TR}^i(x'_i, x_{-i}) \\ &= [U_{TR}(x_i, x_{-i}) - U_{TR}(x_{-i})] - [U_{TR}(x'_i, x_{-i}) - U_{TR}(x_{-i})] \\ &= U_{TR}(x_i, x_{-i}) - U_{TR}(x'_i, x_{-i}). \end{aligned} \quad (33)$$

According to the definition of potential games in Definition 2, taking Eq. (30), as the threat to individual i leads to a potential game with the potential function U_{TR} defined in Eq. (19). The proof is finished. $\square$

Theorem 1. The coalition formation game model of the red team is a potential game with team utility $U_R(X, Y)$ defined in Eq. (19) as the potential function.

Proof. Lemma 1 has proved that the threat to individual i leads to a potential game with the potential function defined by Eq. (24). Therefore, we only need to prove that the individual situation advantage defined in Eq. (29) with the team situation advantage U_{SR} defined in Eq. (19) satisfies the condition of potential games.

Typically, when individual utility equals its marginal contribution to team utility, it results in a potential game with the team utility function serving as the potential function. Therefore, if it can be demonstrated that the individual situation advantage defined in Eq. (29) represents the marginal contribution to the team situation advantage U_{SR} defined in Eq. (19), the above statement is substantiated.

In the following, we will calculate the marginal contribution of i to U_{SR} . From Eq. (19), we have $U_{SR} = S_R(F_R, F_B, X, Y) - S_B(F_R, F_B, X, Y)$. For convenience, $S_R(F_R, F_B, X, Y)$ and $S_B(F_R, F_B, X, Y)$ are denoted by $S_R(R, B)$ and $S_B(R, B)$, respectively. If remove individual i from the red team, we use $S_R(R \setminus \{i\}, B)$ and $S_B(R \setminus \{i\}, B)$ to denote the situation advantage of red team and blue team, respectively.

The total marginal contribution of situation advantage of individual i to the red team denoted by Δ is given by

$$\Delta = \Delta_1 - \Delta_2, \quad (34)$$

where Δ_1 and Δ_2 refer to the marginal contribution of i to the situation advantage of the red team $S_R(R, B)$ and the blue team $S_B(R, B)$, respectively.

Firstly, we calculate the marginal contribution of i to the situation advantage of red team $S_R(R, B)$, which is given by

$$\Delta_1 = S_R(R, B) - S_R(R \setminus \{i\}, B). \quad (35)$$

From Eq. (22), we know that $S_R(R, B) = \sum_{j \in B} S_R^{C_j}$. Without loss of generality, we assume that $i \in C_j$. Obviously, removing i from coalition C_j will only affect the value of $S_R^{C_j}$ while the situation advantage of other coalitions within the red team remains unchanged. Thus, Δ_1 is equivalent to the change in the situation advantage of coalition C_j after removing individual i , i.e.,

$$\Delta_1 = S_R^{C_j} - S_R^{C_j \setminus \{i\}}, \quad (36)$$

where $S_R^{C_j}$ is

$$\begin{aligned} S_R^{C_j} = & x_{1,j} F_{1,j}^r (1 - y_{1,1} F_{1,1}^b) (1 - y_{2,1} F_{2,1}^b) \cdots (1 - y_{m,1} F_{m,1}^b) \\ & + \cdots \\ & + x_{i-1,j} F_{i-1,j}^r (1 - y_{1,i-1} F_{1,i-1}^b) \cdots (1 - y_{m,i-1} F_{m,i-1}^b) \\ & + \boxed{x_{i,j} F_{i,j}^r (1 - y_{1,i} F_{1,i}^b) \cdots (1 - y_{m,i} F_{m,i}^b)} \\ & + x_{i+1,j} F_{i+1,j}^r (1 - y_{1,i+1} F_{1,i+1}^b) \cdots (1 - y_{m,i+1} F_{m,i+1}^b) \\ & + \cdots \\ & + x_{n,j} F_{n,j}^r (1 - y_{1,n} F_{1,n}^b) \cdots (1 - y_{m,n} F_{m,n}^b). \end{aligned} \quad (37)$$

After removing individual i , the boxed formulation of Eq. (37) equals to 0 since $x_{i,j} = 0$. Therefore, the only difference between $S_R^{C_j}$ and $S_R^{C_j \setminus \{i\}}$ is the boxed formulation, and we can get

$$\begin{aligned} \Delta_1 = & x_{i,j} F_{i,j}^r (1 - y_{1,i} F_{1,i}^b) \cdots (1 - y_{m,i} F_{m,i}^b) \\ = & F_{i,j}^r (1 - y_{1,i} F_{1,i}^b) \cdots (1 - y_{m,i} F_{m,i}^b) \\ = & F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b). \end{aligned} \quad (38)$$

Because it is assumed that $i \in C_j$, we have the second formulation in Eq. (38). Considering the generality of the coalition to which individual i belongs, Eq. (38) can be given by

$$\Delta_1 = \sum_{j \in B} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b). \quad (39)$$

Secondly, we calculate the marginal contribution of i to the situation advantage of blue team $S_B(R, B)$, which is given by

$$\Delta_2 = S_B(R, B) - S_B(R \setminus \{i\}, B). \quad (40)$$

Similarly, we have $S_B(R, B) = \sum_{i \in R} S_B^{C_i}$. For any $C_r, r \in R$, we can calculate its situation advantage as follows:

$$\begin{aligned} S_B^{C_r} = & \sum_{j \in B} y_{j,r} F_{j,r}^b \prod_{i \in R} (1 - x_{i,j} F_{i,j}^r) \\ = & y_{1,r} F_{1,r}^b (1 - x_{1,1} F_{1,1}^r) \cdots (1 - x_{i,1} F_{i,1}^r) \cdots \\ & (1 - x_{n,1} F_{n,1}^r) + \cdots + y_{b,r} F_{b,r}^b (1 - x_{1,b} F_{1,b}^r) \cdots \\ & \boxed{(1 - x_{i,b} F_{i,b}^r)} \cdots (1 - x_{n,b} F_{n,b}^r) + \cdots + y_{m,r} F_{m,r}^b \\ & (1 - x_{1,m} F_{1,m}^r) \cdots (1 - x_{i,m} F_{i,m}^r) \cdots (1 - x_{n,m} F_{n,m}^r). \end{aligned} \quad (41)$$

Assuming that i select b from the blue team as its target, i.e., $x_{i,b} = 1$. If removing i from the red team, then we have $x_{i,b} = 0$ and $(1 - x_{i,1} F_{i,1}^r), \dots, (1 - x_{i,b-1} F_{i,b-1}^r), (1 - x_{i,b+1} F_{i,b+1}^r), \dots, (1 - x_{i,m} F_{i,m}^r)$ are remains unchanged as $x_{i,j} = 0$ for $j \in B \setminus \{b\}$. Therefore, the difference in $S_B^{C_r}$ resulting from the departure of i is only associated with the boxed formulation in Eq. (41). This difference is denoted by $\Delta_B^{C_r}$, as given by

$$\begin{aligned} \Delta_B^{C_r} = & y_{b,r} F_{b,r}^b (1 - x_{1,b} F_{1,b}^r) \cdots (1 - x_{i,b} F_{i,b}^r) \cdots \\ & (1 - x_{n,b} F_{n,b}^r) - y_{b,r} F_{b,r}^b (1 - x_{1,b} F_{1,b}^r) \cdots \end{aligned}$$

$$\begin{aligned} & 1 \cdots (1 - x_{n,b} F_{n,b}^r) \\ = & \left[y_{b,r} F_{b,r}^b \prod_{r \in R, r \neq i} (1 - x_{r,b} F_{r,b}^r) \right] (-x_{i,b} F_{i,b}^r). \end{aligned} \quad (42)$$

Considering the generality of the coalition to which individual i belongs and its influence on all the blue coalitions, Δ_2^1 is given by

$$\Delta_2^1 = \sum_{b \in B} \sum_{r \in R} y_{b,r} F_{b,r}^b (-x_{i,b} F_{i,b}^r) \prod_{r \in R} (1 - x_{r,b} F_{r,b}^r). \quad (43)$$

Additionally, it is worth noting that $y_{j,i} = 0$ if individual i is removed from the red team. Hence, there is another component of Δ_2 , denoted by Δ_2^2 . It is easy to conclude that

$$\Delta_2^2 = \sum_{j \in B} y_{j,i} F_{j,i}^b \prod_{k \in R} (1 - x_{k,j} F_{k,j}^r). \quad (44)$$

Combining Eqs. (43) and (45), we can get

$$\Delta_2 = \Delta_2^1 + \Delta_2^2. \quad (45)$$

Substituting Eq. (39), Eq. (43), and Eq. (45) into Eq. (34), yields

$$\begin{aligned} \Delta = & \sum_{j \in B} x_{i,j} F_{i,j}^r \prod_{k \in B} (1 - y_{k,i} F_{k,i}^b) \\ & + \sum_{b \in B} \sum_{r \in R} y_{b,r} F_{b,r}^b x_{i,b} F_{i,b}^r \prod_{r \in R, r \neq i} (1 - x_{r,b} F_{r,b}^r) \\ & - \sum_{j \in B} y_{j,i} F_{j,i}^b \prod_{k \in R} (1 - x_{k,j} F_{k,j}^r). \end{aligned} \quad (46)$$

Obviously, we can conclude that $\Delta = U_{SR}^i$, meaning the individual situation advantage equals the marginal contribution to the red team. Thus, the proof is complete. $\square$

Lemma 2. The coalition formation game model of the blue team is a potential game with team utility $-U_R(X, Y)$ as the potential function.

Proof. Similar to the proof in Theorem 1, it can be proved that the coalition formation game model of the blue team is also a potential game. $\square$

It is seen that the difference in utility function due to the individual unilateral change is exactly the same with the difference in potential function. In this condition, every UAV of each team seeks to optimize its own utility selfishly, leading to an improvement in the overall performance of its team benefiting from the unique properties of potential games.

6. Intra-coalition and inter-coalition iterative algorithm

In a multi-on-multi scenario, the coalition formation problem comprises two sub-problems. Initially, we fix the opponent's coalition structure and update the intra-coalition structure of our team. Subsequently, given the updated coalition structure of our team, we update the opponent's coalition structure, i.e., the inter-coalition structure. Through such iterative cycles, the optimal coalition structures for both teams can be achieved once the termination conditions of the iteration are satisfied. Therefore, we introduce the intra-coalition and inter-coalition iterative algorithm based on this process to solve for the optimal coalition structures for both teams. This algorithm is hereafter abbreviated as the Double-IC algorithm.

6.1. Intra-coalition: iterated best response algorithm

For the intra-coalition sub-problem, Theorem 1 and Lemma 2 establish that the intra-coalition formation games for both sides are potential games. Since the Iterated Best Response (IBR) algorithm guarantees convergence to a Nash equilibrium in finite potential games (Nisan et al., 2007), we adopt this algorithm to solve both intra-coalition formation games. The algorithm iteratively computes each UAV's best response strategy, as formally defined in Definition 9.

Algorithm 1: Iterated Best Response Algorithm

```

1: Input: Decision matrix  $M$ , Opponent decision matrix  $M_{\text{oppo}}$ ,
   threshold  $th$  for algorithm termination
2: Initialize:  $error \leftarrow \infty$ 
3: while  $error > th$  do
4:    $U_T \leftarrow \text{getTeamUtility}(M, M_{\text{oppo}})$ 
5:   for each player  $p$  do
6:     Initialize an empty list  $u_{\text{list}}$ 
7:     for each opponent  $o$  do
8:        $u_i \leftarrow \text{getInUtility}(p, o)$ 
9:       Append  $u_i$  to  $u_{\text{list}}$ 
10:    end for
11:     $br \leftarrow \arg \max(u_{\text{list}})$ 
12:    Update  $M$  by letting player  $p$  choose  $br$ 
13:  end for
14:   $U_{T'} \leftarrow \text{getTeamUtility}(M, M_{\text{oppo}})$ 
15:   $error \leftarrow |U_{T'} - U_T|$ 
16: end while
17: Output: Updated decision matrix  $M$ 

```

Definition 9 (Best Response Strategy (Narahari, 2014)). Given a strategic form game $\Gamma = \langle N, (S_i), (u_i) \rangle$ and a strategy profile $s_{-i} \in S_{-i}$, we say $s_i \in S_i$ is a best response strategy of player i with respect to s_{-i} if $u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}), \forall s'_i \in S_i$.

The detailed solving process is demonstrated in the pseudo code as Algorithm 1, where the $\text{getTeamUtility}()$ and $\text{getInUtility}()$ are used to calculate the team utility and individual utility, respectively.

Algorithm 1 computes the best response strategy matrix as follows: Given a fixed decision matrix of the opposing team, it evaluates the team utility for each possible action of an individual UAV and identifies the action that yields maximum utility as that individual's best response. This process iterates through all UAVs within the team to generate an updated decision matrix. The algorithm terminates when the difference in team utility between consecutive iterations falls below a threshold $error$, at which point the current decision matrix represents the team's best response strategy matrix.

6.2. Inter-coalition: fictitious play algorithm

For the inter-coalition sub-problem, Definition 4 establishes it as a zero-sum game. To solve this game, we employ the Fictitious Play (FP) algorithm, also known as the Brown-Robinson learning process, which is a well-established method for determining the value of zero-sum games (Berger, 2007). This iterative learning process enables each player to estimate and adapt to the opponent's action probability distribution (Bopardikar et al., 2017; Robinson, 1951).

FP is a dynamic process where each player in a strategic game updates their strategy by learning the average strategy of the opponents. In each iteration, a player chooses the best response against the opponents' average strategy. The detailed process of the FP is included in Algorithm 2. In Algorithm 2, function $\text{updateMat}()$ is used to update the average strategy matrix, and $\text{getCoalitions}()$ is used to generate the corresponding coalition structure of strategy matrix.

To evaluate the efficiency of the Double-IC Algorithm, we analyze its time complexity in relation to the number of participants and iterations. The time complexity of the Double-IC algorithm is $O(T' \cdot T \cdot (NM + N^2))$, where T' is the maximum number of iterations of the outer loop, T is the number of iterations of the Algorithm 1.

6.3. Strategic initialization method of Double-IC algorithm

While the initial strategy matrices of both UAV teams were provided as algorithm inputs in the previous section, here we propose a Binary

Algorithm 2: Double-IC Algorithm

```

1: Input: The historical average strategy matrix of the red team and
   the blue team  $M_R, M_B$ , threshold  $th'$  for algorithm termination
2: Initialize:  $error \leftarrow \infty$ 
3: while  $error > th'$  do
4:    $U_R \leftarrow \text{getTeamUtility}(M_R, M_B)$ 
5:    $U_B \leftarrow \text{getTeamUtility}(M_B, M_R)$ 
6:   for  $i$  in  $\{B, R\}$  do
7:     The opponent  $-i$  adopts the historical average strategy matrix
        $M_{-i}$ 
8:     Get the best response matrix  $M_{br}$  by Algorithm 1
9:      $M_i \leftarrow \text{updateMat}(i, M_{br})$ 
10:  end for
11:   $U'_R \leftarrow \text{getTeamUtility}(M_R, M_B)$ 
12:   $U'_B \leftarrow \text{getTeamUtility}(M_B, M_R)$ 
13:   $error \leftarrow \max(|U'_R - U_R|, |U'_B - U_B|)$ 
14: end while
15:  $C_R \leftarrow \text{getCoalitions}(M_R)$ 
16:  $C_B \leftarrow \text{getCoalitions}(M_B)$ 
17: Output: Optimal coalition structure of two teams  $C_R, C_B$ 

```

Table 3

Computational time comparison under different swarm size.

	N	M	Size	ELP	Double-IC
case 1	2	2	16	0.003 s	0.005 s
case 2	3	3	729	0.260 s	0.060 s
case 3	3	4	5184	2.207 s	0.115 s
case 4	4	4	65536	35.105 s	0.169 s
case 5	4	5	6.400×10^5	499.861 s	0.197 s
case 6	5	5	9.765×10^6	7556.461 s	0.550 s
case 7	5	6	1.215×10^8	--	0.720 s
case 8	7	8	1.208×10^{13}	--	1.790 s
case 9	9	10	3.486×10^{18}	--	7.256 s
case 10	10	13	1.378×10^{24}	--	14.091 s

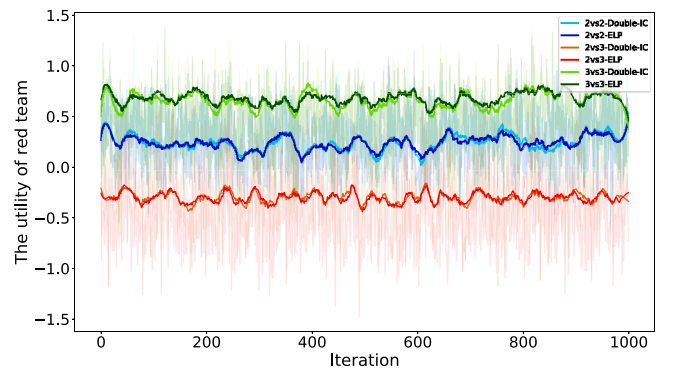


Fig. 3. Utility comparison curves between ELP algorithm and our algorithm.

Integer Programming (BIP) based initialization method to enhance algorithm stability. The initial strategy for the red UAV team can be formulated as the following programming problem:

$$\begin{aligned}
 \max \quad & \alpha \sum_{i \in R} \sum_{j \in B} F_{ij} x_{ij} + (1 - \alpha) \sum_{j \in B} \frac{z_j}{|B|} \\
 \text{s.t.} \quad & \sum_{j \in B} x_{ij} = 1, \\
 & z_j \leq \sum_{i \in R} x_{ij} \leq |R| z_j, \\
 & x_{ij}, z_j \in \{0, 1\}, \quad \forall i \in R, \forall j \in B,
 \end{aligned} \tag{47}$$

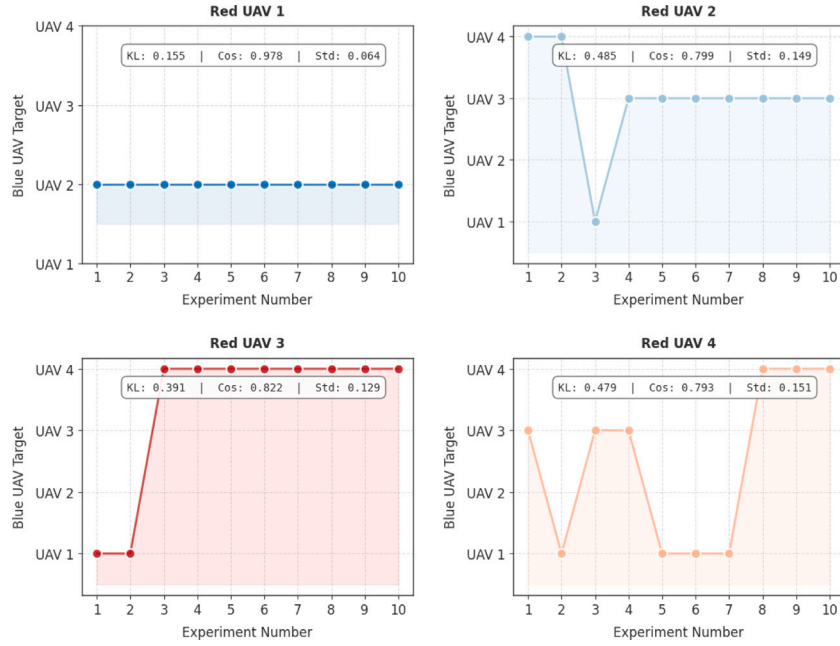


Fig. 4. The results obtained by our method with 10 initial strategies in a 4vs4 scenario.

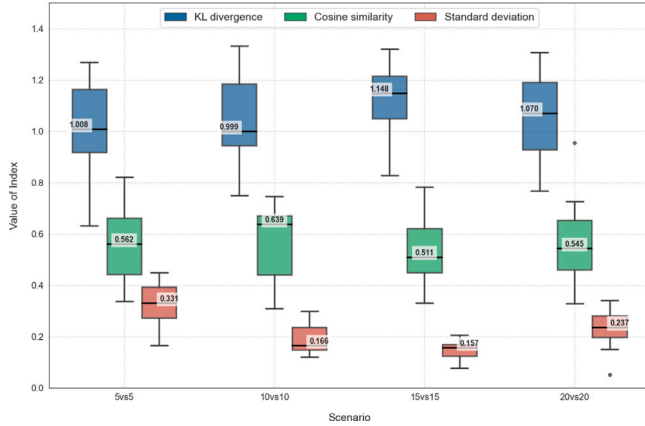


Fig. 5. 3 metrics of results with different initialization strategies in 4 scenarios.

where F_{ij} represents the situation advantage of red UAV i against blue UAV j , x_{ij} is the binary decision variable indicating whether red UAV i selects blue UAV j as its target; z_j serves as an auxiliary variable indicating whether blue UAV j is targeted by any red UAV; and $\alpha \in [0, 1]$ is the weighting coefficient balancing situation advantage and coverage objectives. The first constraint ensures each red UAV selects exactly one target. The second constraint set establishes the logical coupling between target selection variables x_{ij} and coverage indicators z_j , ensuring that z_j equals 1 if and only if at least one red UAV selects blue UAV j as its target.

BIP initialization determines initial target assignments based solely on situation advantages and coverage objectives, independent of the opponent's strategy. This approach enables practical applications where complete information about opponent strategies may be unavailable. The target assignment process can then be iteratively simulated through this strategy-independent BIP initialization followed by the Double-IC algorithm to compute efficient target assignment strategies.

7. Simulation and analysis

This section presents a series of comprehensive simulation experiments to validate the effectiveness and efficiency of our method from multiple perspectives, including: (1) computational performance, (2) algorithm robustness, and (3) applicability in dynamic environments. All simulations were conducted on a personal computer with an AMD Ryzen 7 6800H CPU, 32 GB of RAM, and an NVIDIA GeForce RTX 3080 graphics card.

7.1. Computational performance analysis

7.1.1. Computational time comparison

In this subsection, we compare the computational time required to solve for the exact equilibrium using the ELP algorithm and the approximate equilibrium using the Double-IC algorithm.

Table 3 presents a comparative analysis across 10 scenarios with varying swarm sizes, where N and M denote the number of UAVs for the red and blue teams, respectively. The *Size* column indicates the total number of strategy profiles, calculated as the product of each team's available strategies. The results demonstrate that while the computational time for the ELP algorithm increases exponentially with swarm size due to its exhaustive search nature, our proposed algorithm maintains consistent efficiency. For example, at $N = 4$ and $M = 5$, the ELP algorithm requires approximately 500 s to determine the exact Nash equilibrium, whereas our algorithm achieves an approximate equilibrium in merely 0.197 s. Notably, for scenarios where $N, M > 5$, the ELP algorithm becomes computationally intractable, while our approach continues to provide efficient solutions.

7.1.2. Accuracy validation

In this subsection, we use the ELP algorithm as a benchmark, leveraging its high-precision characteristics, to evaluate the accuracy of the Double-IC algorithm across various scenarios. Specifically, we examine three swarm sizes: 2 vs 2, 2 vs 3, and 3 vs 3, and assess the red team's utility over 1,000 trials. In each trial, the positions, angles, and speeds of the UAVs are randomly generated within their respective ranges.

Fig. 3 presents the team utility comparison of the red team under different algorithms and swarm sizes. The background light-colored

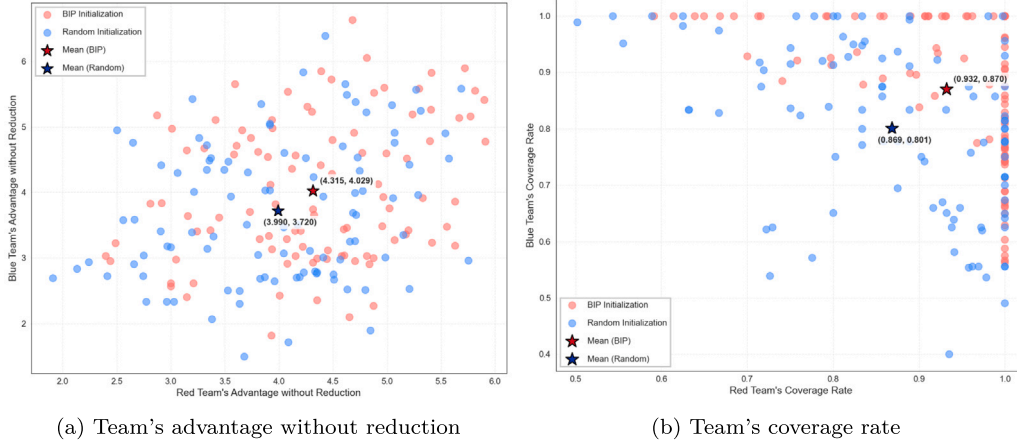


Fig. 6. Performance comparison between BIP-based and random initialization.

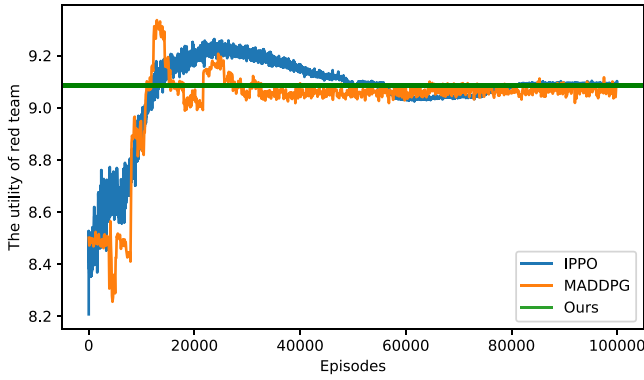


Fig. 7. Comparison between MARL algorithms and our algorithm.

curves show the raw utility values, while the overlaid curves represent the Savitzky–Golay (SG) filtered results for three swarm-size scenarios, each comparing two algorithms. The filtered curves demonstrate highly consistent trends within each scenario, with only minor variations in utility values. Quantitative analysis over 1000 trials reveals average errors of 0.0944, 0.0716, and 0.1220 for the 2 vs 2, 2 vs 3, and 3 vs 3 scenarios, respectively. These results indicate that our algorithm achieves near-equilibrium solutions with minimal error compared to exact equilibrium, while maintaining computational efficiency.

7.2. Robustness and sensitivity analysis

This section investigates the robustness of Double-IC from two aspects: (1) analyzing the sensitivity to initial assignment strategies and validating the effectiveness of the proposed BIP initialization, and (2) comparing with state-of-the-art learning-based methods to show the algorithm's reliability.

7.2.1. Sensitivity analysis of initial assignment strategy

To evaluate the diversity of terminal assignment strategies obtained by the Double-IC algorithm under different initializations, we first introduce three metrics. Let $\mathbf{p} = (p_1, \dots, p_m)$ and $\mathbf{q} = (q_1, \dots, q_m)$ denote two normalized terminal assignment strategies of a UAV, where p_i and q_i represent the probability of choosing target i , and m is the number of potential targets.

- **KL Divergence:** Measures the difference between probability distributions of terminal assignment strategies:

$$D_{KL}(\mathbf{p}||\mathbf{q}) = \sum_{i=1}^m p_i \log \frac{p_i}{q_i}. \quad (48)$$

- **Cosine Similarity:** Evaluates the similarity between terminal assignment vectors:

$$C_{sim}(\mathbf{p}, \mathbf{q}) = \frac{\mathbf{p} \cdot \mathbf{q}}{\|\mathbf{p}\| \|\mathbf{q}\|}. \quad (49)$$

- **Standard Deviation:** Reflects the dispersion of terminal assignment probabilities:

$$\sigma(\mathbf{p}) = \sqrt{\frac{1}{m} \sum_{i=1}^m (p_i - \bar{p})^2}, \quad (50)$$

where $\bar{p} = \frac{1}{m} \sum_{i=1}^m p_i$ is the mean value of strategy $\mathbf{p}$.

In the first experiment, we conduct a detailed analysis in a 4-vs-4 confrontation scenario. We execute the Double-IC algorithm with 10 distinct random initial strategies, Fig. 4 presents these metrics along with the evolution of assignment strategies. As shown in this figure, although the final strategy of the first red UAV remains unchanged, the results of the other three UAVs exhibit differences. Among them, the strategy of the fourth red UAV shows the most significant variation.

The second experiment extends to more complex scenarios, including 5-vs-5, 10-vs-10, 15-vs-15, and 20-vs-20 configurations. For each configuration, we randomly generate 20 different scenarios. Within each scenario, 100 random initial strategies are tested. We calculate the average values of the three metrics for all UAVs under each strategy.

Fig. 5 presents the statistical distribution of performance metrics across varying confrontation scales using box plots. The results demonstrate that the algorithm's performance is significantly influenced by the initial strategy.

Subsequently, we compared the performance of the proposed BIP initialization with that of the random initialization method, using target coverage and team advantage(without reduction) as metrics, across 100 randomly generated target assignment scenarios. As shown in Figs. 6(a) and 6(b), when the Double-IC algorithm is initialized with BIP, both the red and blue teams achieve higher team advantages (4.315, 4.029) and target coverage (0.933, 0.873). In contrast, the random initialization results in lower team advantages (3.990, 3.720) and target coverage (0.872, 0.804).

These results demonstrate that the BIP initialization addresses the algorithm's sensitivity to initial strategies and further validate its effectiveness.

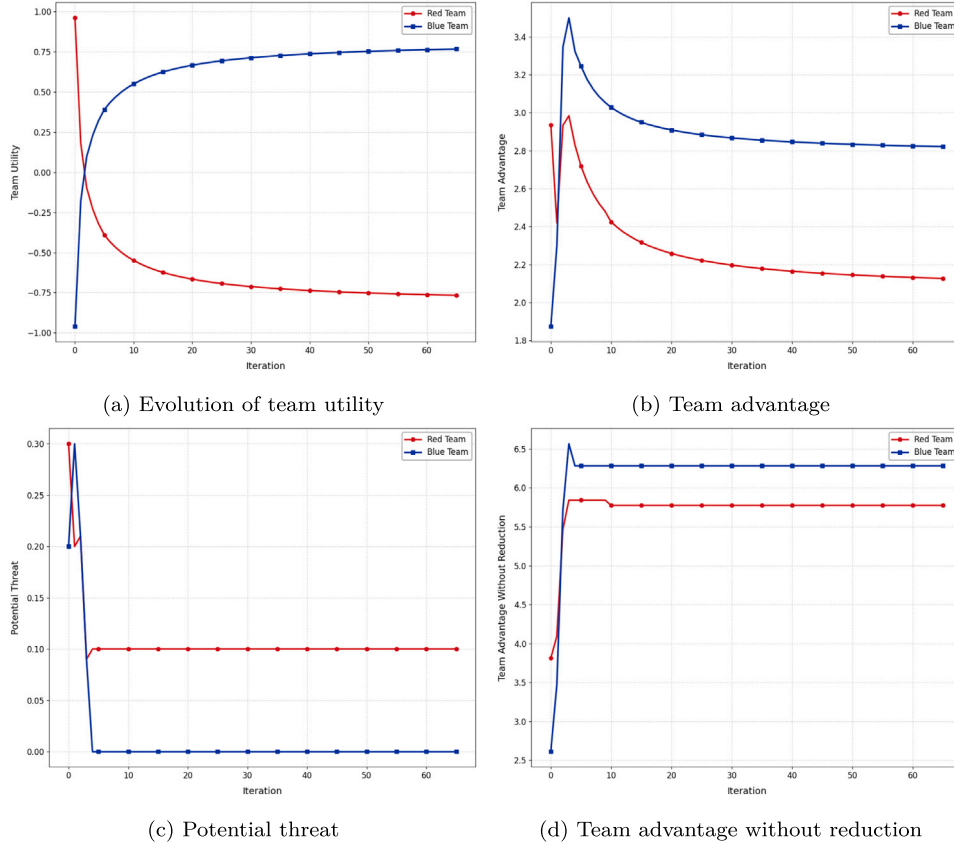


Fig. 8. Evolution of 4 metrics during our method execution in a 10-vs-10 scenario.

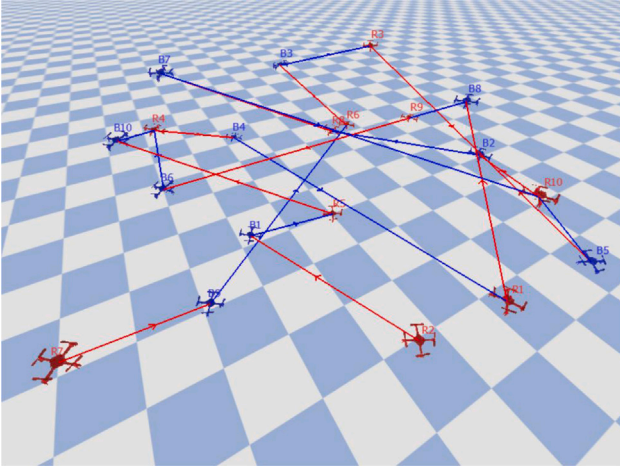


Fig. 9. Target assignment configuration in 10-vs-10 UAV engagement: results from gym-pybullet-drones simulation.

7.2.2. Comparison with learning-based methods

To further validate the robustness of Double-IC, we perform comprehensive comparisons with state-of-the-art reinforcement learning algorithms, including MADDPG (Multi-Agent Deep Deterministic Policy Gradient) and IPPO (Independent Proximal Policy Optimization).

In this simulation, the red team consists of $N = 13$ agents, while the blue team comprises $M = 10$ agents. The initial positions, angles, speeds were randomly generated. Both MADDPG and IPPO are evaluated over 100,000 episodes. For IPPO, the actor and critic learning rates are set to $l_r = 0.00001$ and $l_c = 0.0001$, respectively, with a discount factor of $\gamma = 0.99$ and a hidden network dimension of 64. For MADDPG,

the actor and critic learning rates are set to $l_r = 0.01$, with $\gamma = 0.99$, a clip parameter $\tau = 0.01$, a batch size of 1024, a buffer size of 10,000.

Fig. 7 illustrates the learning curves of the red team's return (team utility). Both the blue and orange curves decline during training due to the zero-sum nature of the red and blue teams, where the red team seeks to maximize utility while the blue team aims to minimize it. These curves ultimately converge to around 9.1, nearly matching the team utility achieved by our algorithm, demonstrating the consistency of our results with those of MARL algorithms and confirming the accuracy of our approach. Notably, while IPPO and MADDPG converge after about 15 h of training (80,000 episodes), our algorithm reaches convergence in approximately 14 s, highlighting its significantly lower computational overhead.

7.3. Dynamic environment validation

To validate the practicability of our method in realistic scenarios, we conduct experiments in the gym-pybullet-drones environment (Panerati et al., 2021), which provides high-fidelity physics simulation and sophisticated UAV dynamics modeling. This section presents two experiments: single-step assignment performance analysis and dynamic confrontation evaluation.

7.3.1. Single-step assignment performance

Within the gym-pybullet-drones simulation environment, we implement the Double-IC algorithm to solve the single-step assignment in a 10-vs-10 UAV engagement scenario. The analysis examines the temporal evolution of 4 metrics for both adversarial teams.

Fig. 8(a) illustrates the team utility for both sides, exhibiting a zero-sum nature. In Fig. 8(b), the team advantage for both sides initially increases and then decreases. Subsequently, in Fig. 8(d), the team advantage without reduction defined in Eq. (21) for both sides shows a

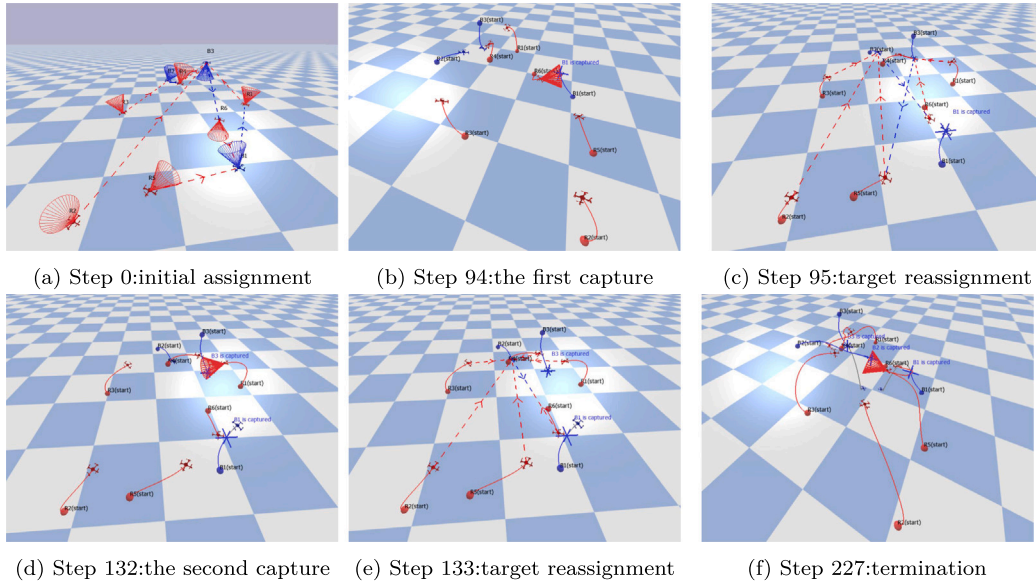


Fig. 10. Dynamic engagement process of UAV swarms in 6-vs-3 scenario.

significant increase, indicating that during the execution of the Double-IC algorithm, both teams achieve higher team advantages, with the reductions against the opponent gradually increasing. Furthermore, Fig. 8(c) demonstrates a decrease in the potential threats for both sides, further confirming that the Double-IC algorithm enhances the decision-making quality of the UAV swarm.

Fig. 9 presents the assignment results computed by the Double-IC algorithm in the gym-pybullet-drones environment.

7.3.2. Dynamic scenario evaluation

This section validates the effectiveness of the Double-IC algorithm in dynamic target assignment scenarios. All UAVs maintain constant velocities during the engagement process. Following initial target assignment, UAVs execute motion control strategies as defined in Chapter 3. When a target enters a UAV's capture zone, the UAV performs capture operations. Additionally, the changes in coalition structures trigger real-time target reassignment through the Double-IC algorithm, ensuring dynamic adaptation throughout the engagement process.

Fig. 10 illustrates the engagement process between two UAV swarms. In the figure, dashed lines represent target assignment connections, solid lines depict motion trajectories, and conical zones indicate capture zones of UAVs. Captured UAVs are explicitly marked on the diagram. Specifically, Fig. 10(a) demonstrates the initial assignment. In Fig. 10(b), the blue UAV B1 is captured by the red UAV R6 at step 94, triggering a dynamic target reassignment as shown in Fig. 10(c). Subsequently, Fig. 10(d) shows that the blue UAV B3 is captured by the red UAV R4 at step 132, initiating another reassignment process in Fig. 10(e), where all remaining red UAVs are assigned to the last surviving blue UAV B1. The simulation terminates in Fig. 10(f) with the encirclement and capture of the blue UAV B1 by the red swarm at step 227.

The second configuration comprises comparative experiments with two variants under identical scenarios: (1) Red team employing Double-IC algorithm versus blue team using BIP initialization strategy, and (2) both teams utilizing BIP initialization strategy. Both experiments are conducted in the same settings with 100 simulation runs, where each run features identical randomized UAV quantities and initial distributions, lasting up to 300 time steps. The performance is evaluated through damage rates and target coverage rate for both sides.

Fig. 11 illustrates the changes in the UAV swarm damage rate and target coverage during a 300-step confrontation under two experimental scenarios. From the figure, it can be observed that when both the

red and blue teams adopt strategies initialized using the BIP approach as their final strategy, they exhibit similar damage rates (around 50%) and target coverage rates (around 75%). When the red team utilizes the Double-IC algorithm proposed in this paper, Fig. 11(b) shows that the red team achieves a significantly higher target coverage rate from the outset of the confrontation and maintains this advantage, with an average target coverage rate of approximately 90% by the end of the confrontation. Fig. 11(a) demonstrates that, during the subsequent confrontation process, the red team maintains a relatively low damage rate (not exceeding 40%), while forcing the blue team's damage rate to increase significantly (ultimately exceeding 60%) and its target coverage rate to decrease (below 70%). This validates the effectiveness of our approach in dynamic scenarios.

8. Conclusion

This paper presents a two-level coalition formation game model for police UAVs to manage aerial threats. The Double-IC algorithm, based on Iterated Best Response and Fictitious Play, is developed to solve the optimal coalition structure. Numerical experiments demonstrate the algorithm's computational efficiency and stability through comparisons with the ELP algorithms. Additionally, the robustness of our method is confirmed via comprehensive sensitivity analyses and benchmarking against MARL algorithms. Finally, the practical applicability of the Double-IC algorithm is demonstrated through its deployment in dynamic target assignment scenarios within UAV swarm operations.

The proposed two-level coalition formation game framework demonstrates significant practical value in scenarios where internal cooperation and external competition coexist. In urban counter-terrorism operations, we can deploy multiple UAV teams to counter illegal drone swarms, where teams require internal coalition formation for surveillance and interception while competing strategically for optimal encirclement. In border security applications, defensive UAV swarms protect against hostile drone infiltration through adaptive internal coalitions for reconnaissance and interception, while engaging in competitive interaction with offensive swarms. The framework also can be applied to military air combat scenarios, where UAV formations engage enemy aircraft groups through internal coalitions for specialized combat roles while maintaining strategic competition at the formation level.

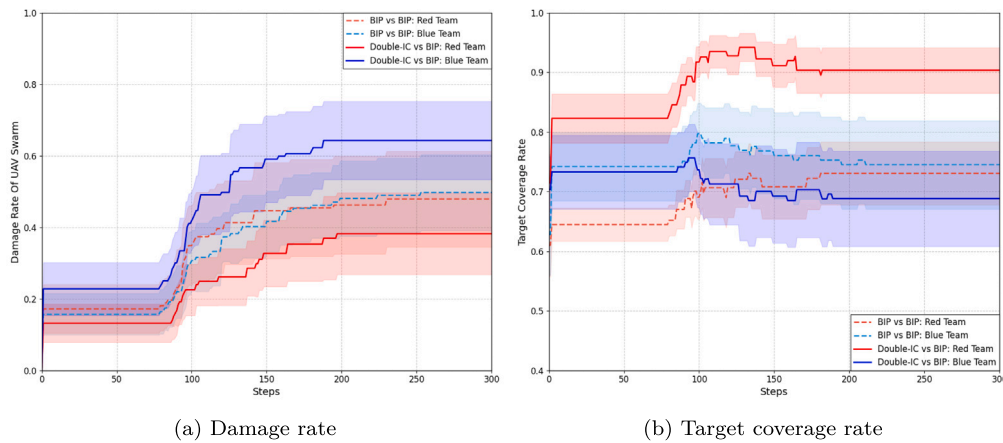


Fig. 11. Comparative analysis of Double-IC algorithm and BIP method in dynamic scenarios.

For future work, several promising directions can be explored. First, while this study focuses on homogeneous UAV swarms, extending the framework to heterogeneous UAV swarms would enhance its practical applicability. Second, the current two-level game framework, which investigates the interaction between intra-team and inter-team strategies, could be enhanced by incorporating environmental noise, communication constraints, limited computational resources, and external threats to improve the robustness of our approach. Furthermore, future research could focus on swarm re-identification and reintegration technologies to address contingencies such as temporary UAV loss. Additionally, regarding dynamic target allocation, long-term cumulative utility optimization could be implemented to replace the current single-step optimization method, thereby achieving far-sighted strategies.

CRedit authorship contribution statement

Zekun Duan: Writing – original draft, Writing – review & editing, Methodology, Conceptualization, Validation. **Genjiu Xu:** Supervision, Project administration, Resource. **Mengda Ji:** Formal analysis, Conceptualization. **Zesheng Li:** Writing – review & editing, Methodology, Visualization. **Binbin Meng:** Investigation, Resource, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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